

Task No: 11
Date:

CRUD Operations In Graph Databases

Aim: To perform CRUD operations like creating, inserting, querying, finding, deleting operations on graph spaces.

* Create Node with Properties

Properties are the key-value pairs using which a node stores data. You can create a node with properties using the CREATE clause. You need to specify these properties separated by commas within the flower braces {}.

Syntax

Following is the syntax to create a node with properties.

```
CREATE (node: label {key1: value, key2: value, ...})
```

* Returning the Created Node

To verify the creation of the node, type and execute the following query in the dollar prompt.

```
MATCH (n) RETURN n
```

* Creating a Relationship Between the Existing Nodes

You can also create a relationship between the existing node using the MATCH clause

Syntax

Following is the syntax to create a relationship using the MATCH clause

```
MATCH (a: Label of Node 1), (b: Label of Node 2)
```

```
WHERE a.name = "name of node 1" AND b.name = "name of node 2"
```

```
CREATE (a) - [: Relation] -> (b)
```

```
RETURN a, b
```

* Deleting a particular Node

To delete a particular node, you need to specify the details of the node in the place of "n" in the above query.

Syntax

Following is the Syntax to delete a particular node from neo4j using the DELETE clause.

```
MATCH (node:label{properties....})  
DETACH DELETE node
```

Create a graph database for student course registration
Create student and dept node and insert values for properties.

```
create (n:student {sid: "VTU14500",  
sname: "John",  
deptname: "CSE"})
```

OUTPUT

Added 1 label, Created 1 node, set 3 properties, Completed after 232 ms.

```
create (n:student {sid: "VTU14501",  
sname: "Dharsana",  
deptname: "ECE"})
```

OUTPUT

Added 1 label, created 1 node, set 3 properties, completed after 16 ms.

```
create (w:student {sid: "VTU14502",  
sname: "Vijay",  
deptname: "CSE"})
```

OUTPUT

Added 1 label, Created 1 node, Set 3 properties, Completed after 12 ms.

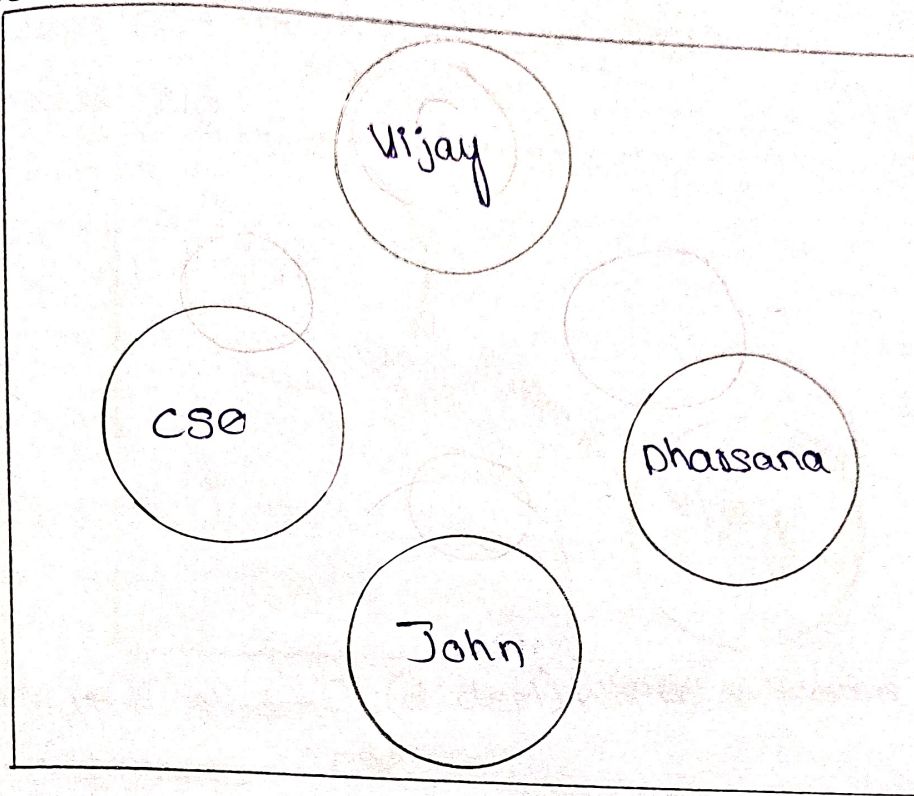
```
create (n:dept {deptname: "CSE", deptid: "d001"})
```

OUTPUT

Added 1 label, Created 1 node, Set 2 properties, completed after 12 ms.

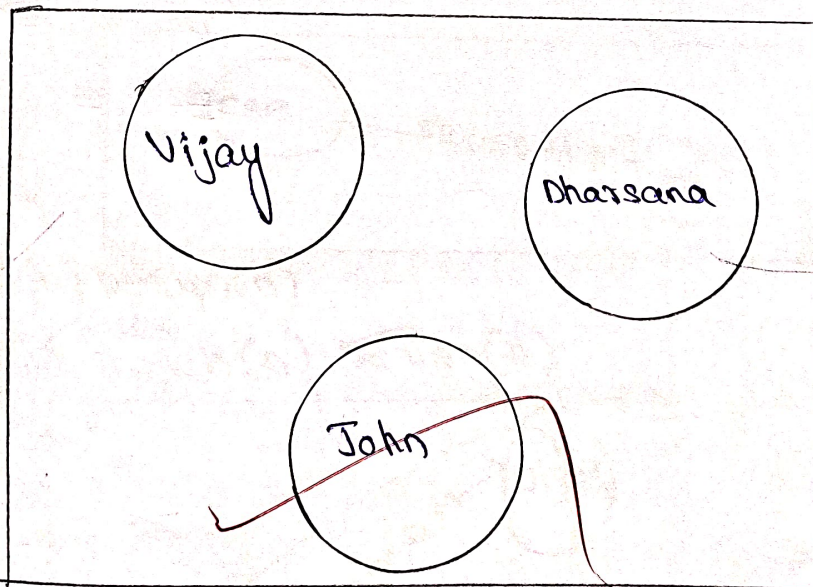
select all the nodes in your database using match command
* match (n) return (n)

OUTPUT



* match (n: student) return (n)

OUTPUT:



a) create relationship between student and cse.

MATCH (S: student), (d: dept) WHERE S.Sname = 'Vijay'

AND d.deptname = 'CSE'

CREATE (S)-[st: STUDIED-AT]-(d)

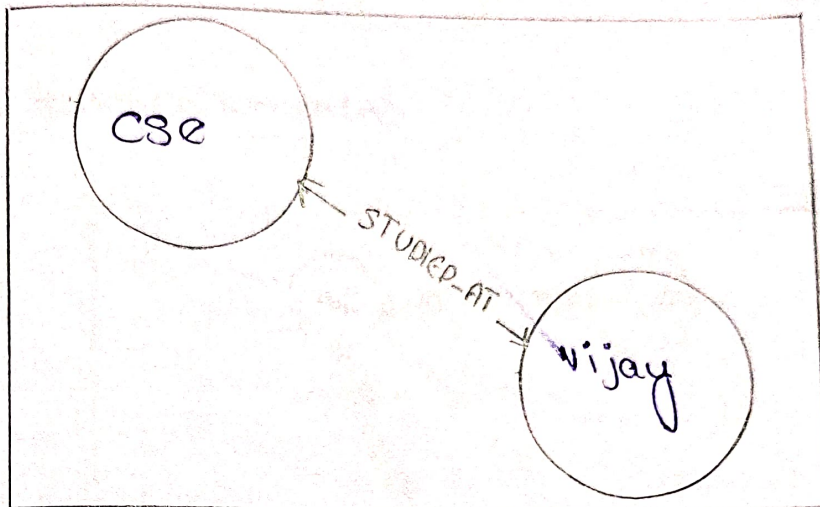
return S, d

OUTPUT

MATCH(S:Student),(d:dept) WHERE S.Sname = 'vijay'. AND
d.depname = 'cse'

CREATE (S) - [st: STUDIED-AT] -> (d)

return: S,d

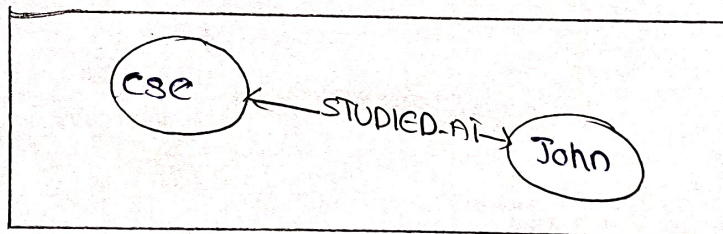


MATCH (S:Student),(d:dept) WHERE S.Sname = 'John' AND
d.depname = 'cse'

CREATE (S) - [st: STUDIED-AT] -> (d)

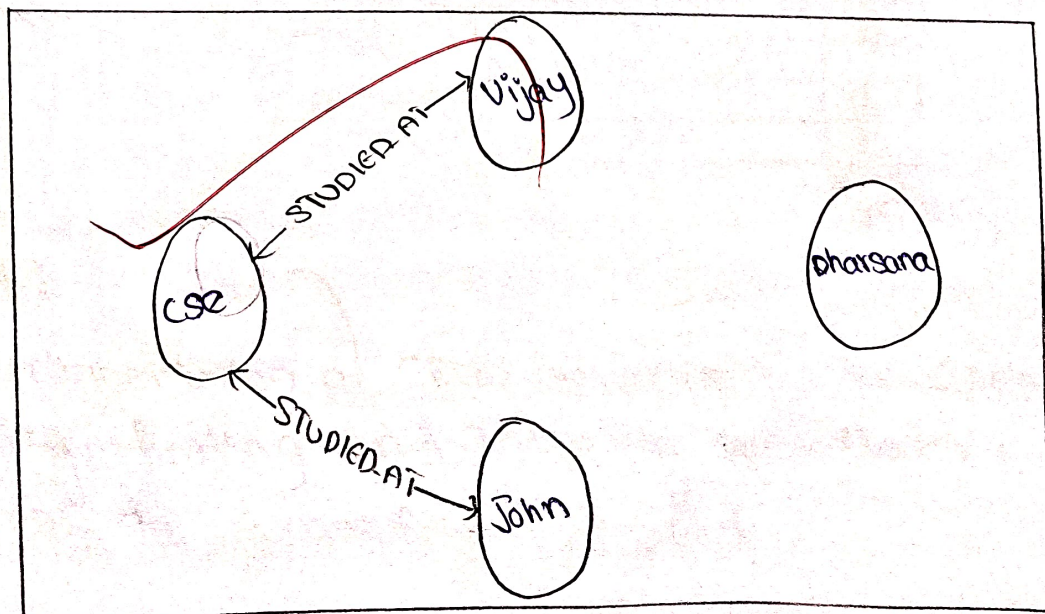
return S,d

OUTPUT



match(n) return(n)

neo4j \$ match(n) return(n)



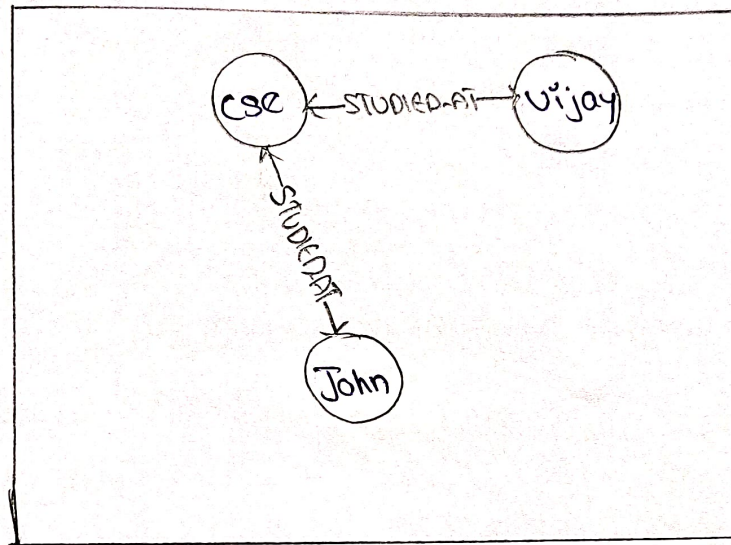
b) Delete a node from Student

match(n: Student {Sname: 'Dhassana'}) DELETE(n)

OUTPUT:

Deleted 1 node, Completed after 10834 ms.

neo4j\$ match(n) return (n)



VEL TECH	
EX NO.	11
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	4
RECORD (5)	
TOTAL (20)	14
SIGN WITH DATE	15/10

Result:

The implementation of CRUD operations like Creating, Inserting, finding and removing operations using GraphDB is successfully executed.